Critical Infrastructure Security

IY5612 (IY3612/IY4612) — Academic Year 2025/2026 (Unit 3)

Stephen D. Wolthusen
stephen.wolthusen@rhul.ac.uk

Department of Information Security and Communication Technology
Norwegian University of Science and Technology, Norway

and

Department of Information Security
Royal Holloway, University of London, United Kingdom

Unit 3— Networks and Dependencies

Contents

Models of Large-Scale Networks

Models of Large-Scale Networks
- Introduction
- Graph Theoretic Concepts
- Characterising Graphs
- Random Graphs

Part I

Models of Large-Scale Networks

Network Models
Random and Scale-Free Graphs

This block covers elementary concepts of network models utilised in a large number of areas relevant to cyber security:

- Fundamental Graph Theoretic Concepts
- Methods for Characterising Graphs
- Random Graphs

Recommended Reading

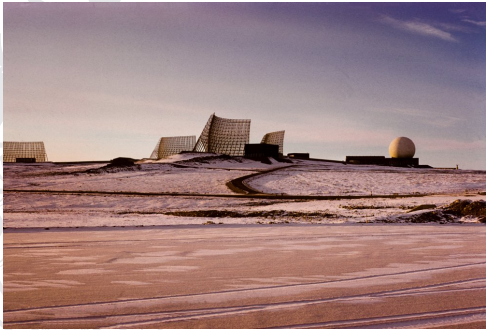
- N. K. Svendsen, S. Wolthusen:
  Modelling Approaches
  In: J. Lopez, R. Setola, S. Wolthusen (eds.):
  Critical Infrastructure Protection: Information Infrastructure Models, Analysis, and Defense
  Lecture Notes in Computer Science Vol. 7130, Springer-Verlag, 2012
- M. van Steen:
  Graph Theory and Complex Networks: An Introduction
  Maarten van Steen (self-published, 2010)
- M. E. J. Newman:
  Networks: An Introduction
  Oxford University Press, 2010

Chapters 1 (Introduction), 6 (Mathematics of Networks)

Why Robust Networks Matter (1/3)

The subsequent block will cover network robustness explicitly, but to give an idea why understanding large networks matters:

- On 24 November 1961 — at the height of the cold war and during confrontation over Berlin, nuclear war being contemplated by both the US/NATO and the USSR/Warsaw Pact



The US had its early warning (Ballistic Missile Early Warning System, BMEWS) radar systems along the northern hemisphere along the anticipate track of Soviet missiles and bombers, the most important one on Greenland (Thule Air Force Base), another at RAF Fylingdales

Why Robust Networks Matter (2/3)

Incident timeline:

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A **single telephone switch operated by AT&T** in Colorado had failed, shutting down all early warning and voice communication between SAC, the NORAD command post, and the hot line between SAC and NORAD commands

Why Robust Networks Matter (3/3)

Bell System #4A toll crossbar switching system

Source: AT&T Archives and History Centre

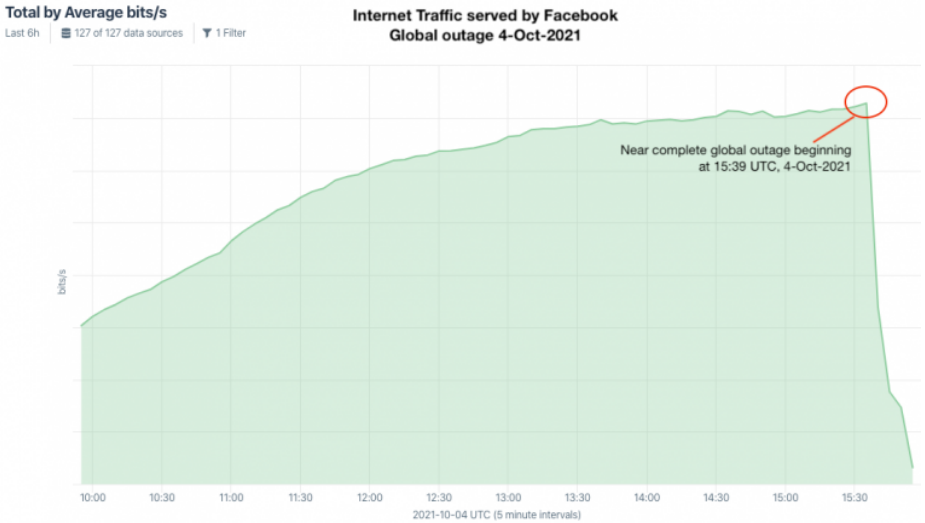
Coda

This was not even the first time that BMEWS had nearly caused World War III:

- Shortly after commissioning, it had detected a simultaneous launch of over 1000 missiles from Siberia with high certainty
- Protocol in place would have allowed the STRATCOM commander to launch a counter-strike without involving the NCA
- Cooler heads prevailed, noting that the Soviet leader Nikita Khruschev was in New York at the UN General Assembly at the time
- Cause for the alert was the moon rising over Siberia

If It’s Not A Single Point of Failure. . .

Issues at Facebook October 4, 2021



Source: Kentik LLC

. . . No Revenue

On 4 October 2021, Facebook (Meta) withdrew all BGP routes to the Internet, making all their applications (including Instagram, WhatsApp) inaccessible

- A configuration error disconnected Meta"s internal backbone network; leading to its custom-designed internal BGP servers to quit advertising routes **including to DNS servers hosted exclusively internally**
- This circular dependency removed all authoritative DNS servers for Meta; theoretically these ran on a separate network, but had advertisements withdrawn when backbone became unreachable

Errors such as BGP blips are fairly frequent, and one does wonder why there is no two-person rule for updating configurations

- Facebook staff could not communicate with internal tools or even enter buildings since access control required access to . . .
- Facebook stock went down 5% in a day (US$ 50bn)

Network Models for Cyber Security?

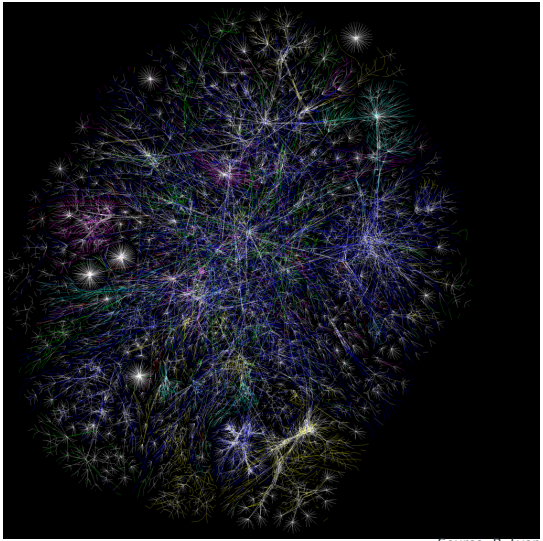
In the context of Cyber Security — particularly where interactions with the C(I)I are concerned — **networks** have a more broad meaning than computer networks

- Flows of resources are a typical example amenable to a network-type model, whether physical (e.g. passengers) or logical (e.g. financial transactions)

Understanding security problems can benefit considerably from choosing the right abstraction level:

- The design of cryptographic protocols tends to use a highly abstract communication model, potentially not capturing interesting aspects
- Studying protocols at the implementation level can obscure effects and behaviour with extraneous detail

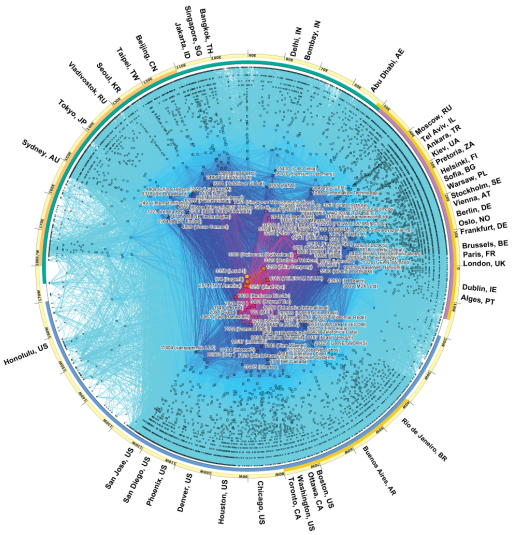
BGP Routes on the Internet (Jan. 2005)



Source: B. Lyon — http://www.opte.org/maps/

BGP Routes on the Internet

IPv4 AS Core (2017)



Source: CAIDA

Representing Networks

We consider arbitrary networks as collections of **nodes** (or **vertices**) and connections (**edges**):

Definition (*Graph*)

A graph G is a 2-tuple (V, E) where V is a set of vertices and E a set of edges. An edge $e \in E$ is said to connect two vertices $u, v \in V$, and is denoted as $e = \langle u, v \rangle$
The vertex set and edge set of G are denoted as $V(G)$, $E(G)$, respectively

Definition (*Neighbour Set*)

For a graph G and vertex $v \in V(G)$, the neighbour set $N(v)$ of v is the set of vertices (other than v) adjacent to v:

$$N(v) = \{w \in V(G) \mid v \neq w, \langle v, w \rangle \in E(G)\}$$

Degrees and Subgraphs

Definition (*Vertex Degree*)

The number of edges incident with a vertex v is called the degree of v, denoted as $\delta(v)$. Loops are counted twice

Definition (*Subgraph*)

H is a subgraph of G if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$ such that $\forall e \in E(H)$ with $e = \langle u, v \rangle : u, v \in V(H)$.

Definition (*Induced Subgraphs*)

A subset of vertices induces a subgraph H:

- *$V^* \subseteq V(G)$ has the vertex set V^* and edge set $\{\langle v, w \rangle \in E(G) | v, w \in V^*\}$: $H = G[V^*]$*

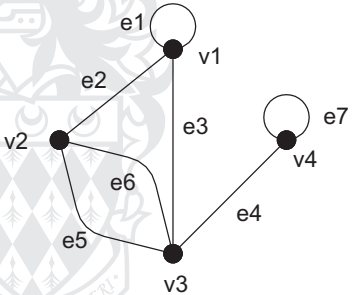
Similarly, a subset of edges induces a subgraph H:

- *$E^* \subseteq E(G)$ has the vertex set $V(G)$ and edge set E^*: $H = G[E^*]$*

Representing Graphs
Adjacency

A graph can be represented in the form of a (symmetric) matrix with edges represented as values in the matrix referred to as the adjacency matrix showing adjacent vertices:

	v_1	v_2	v_3	v_4
v_1	2	1	1	0
v_2	1	0	2	0
v_3	1	2	0	1
v_4	0	0	1	2



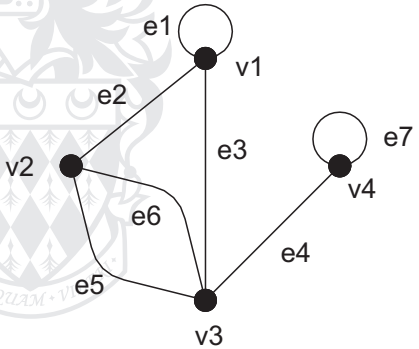
We note that G is simple $\Leftrightarrow$ $\mathbf{A}[i,j] \leq 1$ and $\mathbf{A}[i,i] = 0$

Otherwise, multiple edges in (undirected) graphs can be represented: $\forall v_i$: $\sum_{j=1}^{n} \mathbf{A}[i,j] = \delta(v_i)$

Representing Graphs
Incidence

A matrix with rows representing vertices and columns representing edges is called an incidence matrix:

	e_1	e_2	e_3	e_4	e_5	e_6	e_7
v_1	2	1	1	0	0	0	0
v_2	0	1	0	1	1	0	0
v_3	0	0	1	1	1	1	0
v_4	0	0	0	0	0	1	2

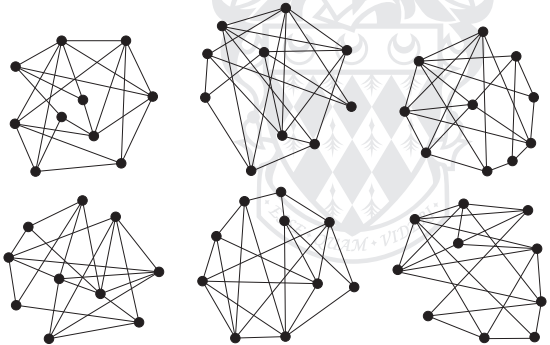


For an undirected graph, G is simple only if $\mathbf{M}[i,j] \leq 1$

Graph Isomorphism

Definition (*Graph Isomorphism*)

G_1 and G_2 are isomorphic if there exists a one-to-one mapping $\phi : V_1 \to V_2$ such that for each edge $e \in E_1$ with $e_1 = \langle v, u \rangle$ there is a unique edge $e_2 \in E_2$ with $e_2 = \langle \phi(v), \phi(u) \rangle$



Connectivity (1/3)

Definition (*Walks, Trails, Paths, and Cycles*)

A (v_0, v_k)-walk is a sequence $v_0 e_1 v_1 e_2 \ldots v_{k-1} e_k v_k$ with $e_i = \langle v_{i-1}, v_i \rangle$

A trail is a walk with distinct edges, and a path is defined as a walk with distinct vertices

A cycle is a trail with distinct vertices except $v_0 = v_k$

Definition (*Connectedness*)

Vertices $u \neq v$ in G are connected if there is a $(u, v) - path$ in G. G is connected if all pairs of distinct vertices are connected

Connectivity (2/3)

Definition (*Component*)

$H \subseteq G$ is a component of G if H is connected and not contained in a connected subgraph of G with more vertices or edges. The number of components of G is $\omega(G)$

Definition (*Vertex and Edge Cuts*)

For a graph G let $V^* \subset V(G)$ and $E^* \subset E(G)$

If $\omega(G - V^*) > \omega(G)$ then V^* is called a vertex cut

If $\omega(G - E^*) > \omega(G)$ then E^* is called an edge cut

$\kappa(G)$ denotes the size of a minimal vertex cut for G, and $\lambda(G)$ the size of a minimal edge cut

Connectivity (3/3)

Definition (*k-Connectivity*)

If $\kappa(G) \geq k$ for some k, then G is called k-connected

Definition (*Vertex and Edge Independence*)

Let $\mathcal{P}(u,v)$ be a collection of paths between vertices u and v; two paths P,Q are called vertex independent if $\forall P,Q \in \mathcal{P}(u,v) : V(P) \cap V(Q) = \{u,v\}$
If $\forall P,Q \in \mathcal{P}(u,v) : E(P) \cap E(Q) = \emptyset$, then P,Q are called edge independent

One basic result helping to understand connectivity is **Menger's theorem**

Menger's Theorem

The theorem covers both edge connectivity and vertex connectivity (the requirement for vertex connectivity is stronger, calling for non-adjacency whilst edge connectivity requires only that u and v be distinct):

Theorem (*Menger*)

Let G be a graph with two nonadjacent vertices u and v

The minimum number of vertices in a vertex cut that disconnects u and v is equal to the maximum number of pairwise vertex-independent paths between u to v

The minimum number of edges in an edge cut that disconnects u and v, is equal to the maximum number of pairwise edge-independent paths betweeen u and v

Menger's Theorem — Corollaries

This gives rise to a number of corollaries, including:

Corollary

- *G is k-connected if and only if any two distinct vertices are connected by at least k vertex-independent paths*
- *G is k-edge connected if and only if any two distinct vertices are connected by at least k edge-independent paths*

Corollary

Each edge of a 2-edge-connected graph lies on a cycle

Directed Graphs

So far we have considered only undirected graphs, but frequently we are interested in properties that are better expressed with directed edges:

Definition (*Directed Graph*)

A directed graph or digraph D is a tuple (V, A) of vertices V, and a collection of arcs A where each arc $a = \langle \overrightarrow{u, v} \rangle$ joins a vertex (tail) $u \in V$ to another (not necessarily distinct) vertex (head) v

The term arc and edge are sometimes used interchangeably, many concepts extend in a straightforward way from undirected graphs

Directed Graphs
Connectivity and Reachability

Definition (*Walks, Trails, and Paths in Directed Graphs*)

A directed $(\mathbf{v_0}, \mathbf{v_k})$*-walk is an alternating sequence* $[v_0, a_0, v_1, a_1, \ldots, v_{k-1}, a_{k-1}, v_k]$ *with* $a_i = \overrightarrow{\langle v_i, v_{i+1} \rangle}$
A directed trail is a directed walk with distinct arcs; a directed path is a directed trail with distinct vertices

Definition (*Connectivity in Directed Graphs*)

D is strongly connected if there exists a directed path between every pair of distinct vertices from D
D is weakly connected if its underlying (undirected) graph is connected

Definition (*Reachability*)

Vertex v is reachable from vertex u if there exists a directed (u, v)*-path*

Weighted Graphs

A number of questions make further information for each edge desirable, for example the capacity of network links when constructing routes

- This can be achieved in different ways with distinct properties: Real-valued, cardinal weights, colours

Definition (*Weighted Graph (Real-Valued)*)

In a weighted graph G each edge e has an associated real-valued weight $w(e) < \infty$

For $H \subseteq G$, $w(H) = \sum_{e \in E(H)} w(e)$.

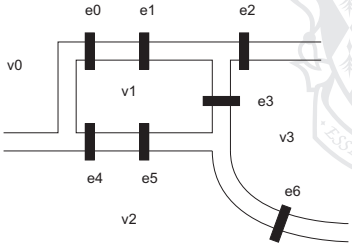
Identifying the shortest path in a graph is a very simple example:

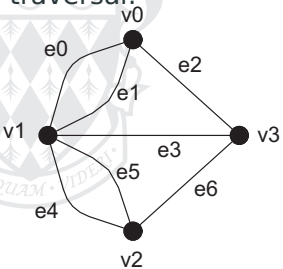
- **Dijkstra's Algorithm** solves the single-source shortest path problem for non-negative path costs, constructing a shortest-path tree

Towards Routing
Eulerian Circuits

The Pregel River flowing through the city of Königsberg contained two islands. Both sides of the mainland were connected one side of the mainland with one island with a third bridge joining the same side of the mainland with the other island and a further bridge between the two islands

Leonhard Euler formalised the problem whether it was possible to cross each bridge exactly once during a single traversal:





Euler Circuit

Definition (*Euler Circuit*)

A circuit of an undirected graph graph G is a (u, v)-walk in which $u = v$ (i.e., it is a closed walk) and that traverses each edge in G

An Euler circuit is a circuit in which all edges are traversed exactly once

Theorem

A connected graph G (with more than one vertex) has an Euler tour if and only if it has no vertices of odd degree

Constructing Eulerian circuits can be achieved in linear time (e.g. using Hierholzer's algorithm)

Hamilton Cycles

We can also ask if we can construct paths that contain every vertex:

Definition

A Hamilton path of a connected graph G is a path that contains every vertex of G
A Hamilton cycle is a cycle containing every vertex of G. G is called Hamiltonian if it has a Hamilton cycle

Note that unlike for Eulerian circuits, there is no efficient algorithm to determine whether a graph is Hamiltonian:

- The Hamiltonian cycle is a special case of the **Travelling Salesman Problem** determining the lowest cost of a cycle over a weighted graph — hence it is $\mathcal{NP}$-complete

Distance Statistics

Particularly for large graphs we may be interested in *statistical* properties:

Definition (*Distance Statistics*)

Let G be a connected graph. Recall that $d(u,v)$ is the distance between vertices u and v: The length of a shortest path between u and v

Eccentricity	$\epsilon(u)$:	$\max\{d(u,v) \mid v \in V(G)\}$
Radius	$rad(G)$:	$\min\{\epsilon(u) \mid u \in V(G)\}$
Diameter	$diam(G)$:	$\max\{d(u,v) \mid u,v \in V(G)\}$

Note that the definitions apply to both directed and undirected graphs, but may differ between the digraph and the underlying graph

Path Lengths

Average and Characteristic Path Lengths

Definition (*Average Path Length*)

Let G be a connected graph; $\bar{d}(u)$ is the average length of shortest paths from u to any other vertex v:

$$\bar{d}(u) = \frac{1}{|V|-1} \sum_{v \in V, v \neq u} d(u,v)$$

The average path length $\bar{d}(G)$:

$$\bar{d}(G) = \frac{1}{|V|} \sum_{u \in V} \bar{d}(u) = \frac{1}{|V|^2 - |V|} \sum_{u,v \in V, u \neq v} d(u,v)$$

Definition (*Characteristic Path Length*)

*The characteristic path length is the **median** over all $\bar{d}(u)$*

Clustering Properties

Clustering and Vertex Clustering Coefficients

An interesting property of graphs (networks) is the way that neighbouring vertices are connected to each other, this is called **clustering**:

Definition (*Vertex Clustering Coefficient*)

Let G be a simple, connected, undirected graph with a vertex $v \in V(G)$ having the neighbour set $N(v)$:

- *Let $n_v = |N(v)|$*
  (max. number of edges between neighbours is $\binom{n_v}{2}$)
- *Let m_v be the number of edges in subgraph induced by $N(v)$:*
  $m_v = |E(G[N(v)])|$

The Clustering Coefficient $cc(v)$ is then defined as

$$cc(v) = \begin{cases} m_v / \binom{n_v}{2} = \frac{2 \cdot m_v}{n_v(n_v - 1)} & \text{if } \delta(v) > 1 \\ \text{undefined} & \text{otherwise} \end{cases}$$

Clustering Properties
Graph Clustering Coefficients

The clustering concept can also be extended from vertices to encompass the entire graph:

Definition (*Graph Clustering Coefficient*)

Let G be a simple, connected, and undirected graph

Let $V^* = \{v \in V(G) | \delta(v) > 1\}$

The Clustering Coefficient $CC(G)$ for G is then

$$CC(G) = \frac{1}{|V^*|} \sum_{v \in V^*} cc(v)$$

Centrality

Centrality Measures and Vertex Centrality

Centrality can be defined based on different metrics, with the intuition being nodes that are the furthest away from the edge of the graph:

Definition (*Graph Centre and Vertex Centrality*)

Let G be a (strongly) connected graph

The centre $C(G)$ is the set of vertices with minimal eccentricity:

$$C(G) = \{v \in V(G) | \epsilon(v) = rad(G)\}$$

The eccentricity-based vertex centrality $c_E(u)$ of u can then be described as

$$c_E(u) = \frac{1}{\epsilon(u)}$$

Centrality
Closeness Centrality

We can also ask how close a given vertex is to **all** other vertices:

Definition (*Closeness Centrality*)

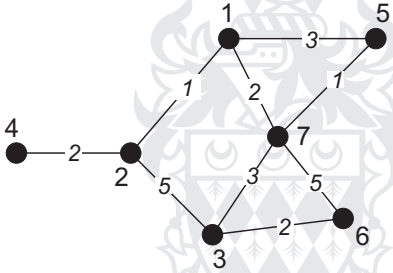
Let G be a (strongly) connected graph. The closeness $c_C(u)$ of u is defined as

$$c_C(u) = \frac{1}{\sum_{v \in V(G)} d(u,v)}$$

Measures such as these (including Betweenness Centrality discussed below as well as *degree*, *shortest-path*, and *Eigenvector centrality*) are utilised extensively in **social network analysis** research, where the precondition of strongly connected underlying graphs can sometimes be problematic in applications

Closeness Centrality
Example Graph

Consider the following weighted graph:



Vertex:	**1**	**2**	**3**	**4**	**5**	**6**	**7**
$\epsilon(u)$	7	7	7	9	6	9	5
$\sum d(u,\cdot)$	21	22	27	32	24	37	29
$c_C(u)$:	0.048	0.045	0.037	0.031	0.042	0.027	0.034

Betweenness Centrality

In addition to outright cut vertices, we can also consider where removal of a vertex would significantly increase distance between other vertices:

Definition (*Betweenness Centrality*)

Let G be a simple and (strongly) connected graph with $S(x,y)$ the set of shortest paths between x and y
Let $S(x,u,y) \subseteq S(x,y)$ be paths that pass through u
The Betweenness Centrality $c_B(u)$ of u is

$$c_B(u) = \sum_{x \neq y} \frac{|S(x,u,y)|}{|S(x,y)|}$$

Intuitively, this could be considered a measure of the *importance* of a vertex, and in networks such nodes would experience higher *load*

Random Graphs

A simple way of modelling networks is as a random graph where edges $\langle u, v \rangle$ between given vertices u, v exist with with probability p:

Definition (*Erdős-Rényi Graph*)

- *Let $ER(n, p)$ be an undirected graph $ER(n, p)$ with n vertices where the edge $\langle u, v \rangle$ $(u \neq v)$ exists with probability p*
- *For given M edges, $ER(n, M)$ is an undirected graph with n vertices and M edges, each incident to randomly chosen pairs of vertices*

Note that $ER(n, p)$ is simple: $ER(n, M)$ is allowed to have loops and multiple edges between the same vertices

Erdős-Rényi Graph Properties

The probability that the degree of u is k is $\mathbb{P}[\delta(u) = k]$

- There are a maximum of $n-1$ other vertices that can be adjacent to u
- We need to choose k other vertices, out of $n-1$, to connect to $u \Rightarrow \binom{n-1}{k}$ possibilities
- Probability of having exactly one specific set of k neighbours hence is

$$p^k(1-p)^{n-1-k}$$

This gives

$$\mathbb{P}[\delta(u) = k] = \binom{n-1}{k} p^k (1-p)^{n-1-k}$$

Erdős-Rényi Graphs
Average Vertex Degree

By definition, all vertices in an Erdős-Rényi graph have the same probability of have degree k, meaning that we can treat the degree distribution as a stochastic variable δ

- We now know that δ follows a binomial distribution

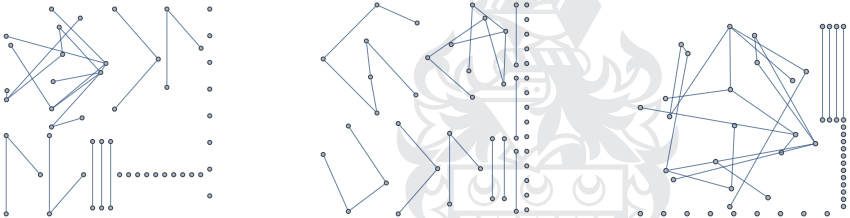
Computing the average (or expected value) of a stochastic variable x requires:

$$\bar{x} = \mathbb{E}[x] = \sum_k k \cdot \mathbb{P}[x = k]$$

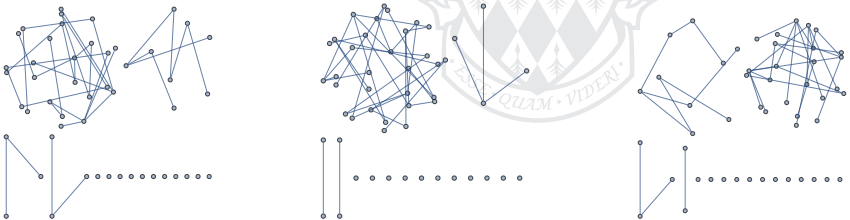
Intuitively, each vertex has a maximum of $n - 1$ incident edges, so we can expect it to have $p(n - 1)$ edges

Examples of Erdős-Rényi Graphs (1)

Random Graphs with $n = 50$ vertices and 25 edges each:



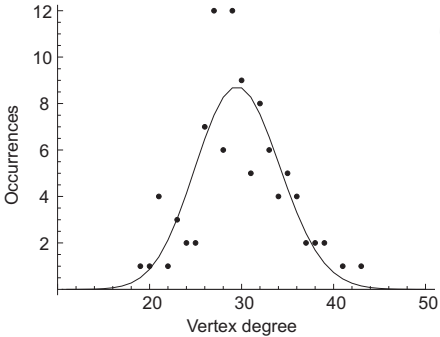
Random Graphs with $n = 50$ vertices and 35 edges each:

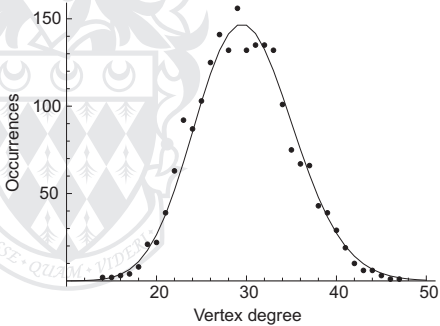


Examples of Erdős-Rényi Graphs (2)

Note

$ER(n,p)$ represents a set of Erdős-Rényi graphs: most $ER(n,p)$ graphs are not isomorphic!





On the preceding slide the number of edges was kept constant, but for $ER(n,p)$ this is obviously not the case

Erdős-Rényi Graph Properties

Random graphs (and particularly ER graphs) have been studied extensively

- These properties include e.g. connectivity and matchings, path properties, colourings, and matchings

As an example, consider the average path length of an ER graph, which can be characterised simply:

Observation

For any large $H \in ER(n,p)$ it can be shown that the average path length $\bar{d}(H)$ is equal to

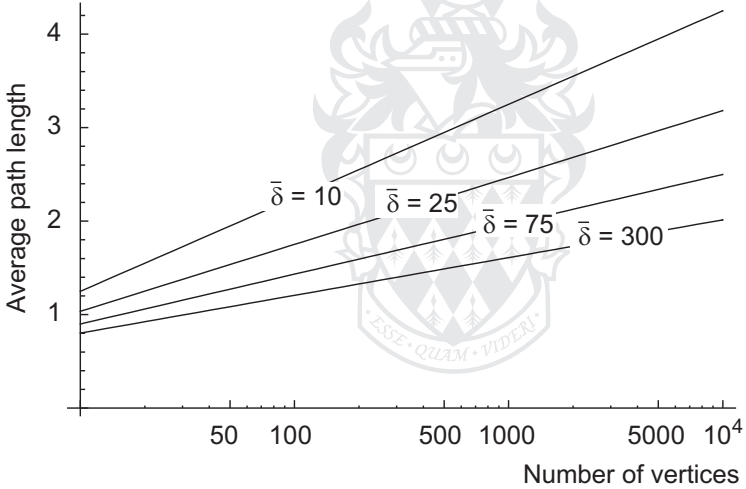
$$\bar{d}(H) = \frac{\ln(n) - \gamma}{\ln(pn)} + 0.5$$

(γ is the the *Euler constant*, ≈ 0.57772)

$$\gamma = \lim_{n\to\infty} \left(\sum_{k=1}^{n} \frac{1}{k} - \ln(n)\right)$$

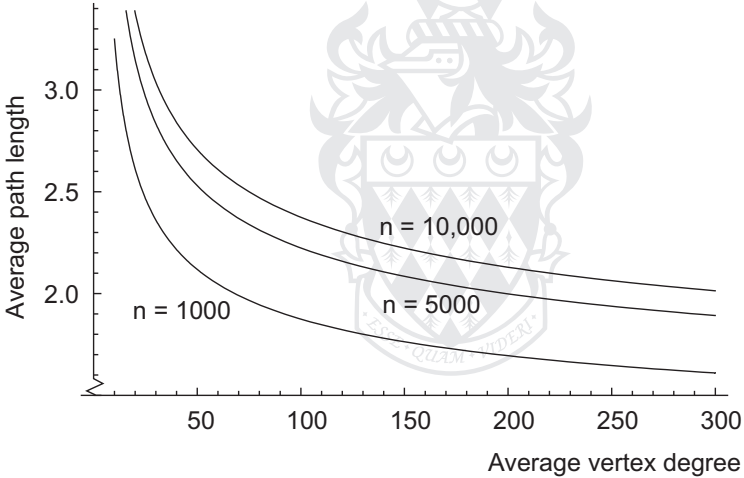
Erdős-Rényi Average Path Length (1)

Consider the average path length where the vertex degree is kept constant, but varying the number of vertices:



Erdős-Rényi Average Path Length (2)

Average path length where the size of the graph is kept constant, but the vertex degree is changed:



Clustering Coefficients

Clustering Coefficients in Erdős-Rényi Graphs

Recall that the clustering coefficient is the fraction of edges between neighbours and the maximum number of possible edges. In an Erdős-Rényi graph we therefore have:

- Expected number of edges between k neighbours:

$$\binom{k}{2} p$$

- Maximum number of edges between k neighbours

$$\binom{k}{2}$$

- Expected clustering coefficient for every vertex:

$$p$$

The transition around $k = 1$ is reminiscent of *phase transitions* observed in physical systems

Summary

This section provided material on

- Introduction to robust networks
- Graph theoretical concepts
- Basic techniques for characterising graphs
- Introduction to random graphs